

ゼミノート (2026/04/10)

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1 Truncated microsupport

Functorial Properties

Proposition 1. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in \mathbf{D}^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(1.1) \quad \text{SS}_k(\mathbf{R}f_* F) \subset f_\pi f_d^{-1}(\text{SS}_k(F)).$$

The equality holds in case f is a closed embedding.

2 Truncated microlocal Morse theory

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

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Proposition 2. Let $G \in \mathbf{D}^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y_t} \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_o a real number.

(i) Assume that, for any $y \in Y \setminus \tilde{Y}_{t_o}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then the following statements hold.

(a) For $t > t_o$, the natural morphism

$$\mathbf{R}^j f_* G \rightarrow \mathbf{R}^j f_{t*} j_t^{-1} G$$

is isomorphic for $j < k$ and injective for $j = k$.

(a') For $t \geq t_o$, the natural morphism

$$\mathbf{R}^j \tilde{f}_* G \rightarrow \mathbf{R}^j \tilde{f}_{t*} \tilde{j}_t^{-1} G$$

is isomorphic for $j < k$ and injective for $j = k$.

(b) We have

$$\begin{aligned} \text{SS}_k(\mathbf{R} f_* G) &\subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ &= f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any $y \in Y \setminus \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$(c) \quad \mathbf{R}^j f_! G \cong \mathbf{R}^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o,$$

$$(c_k) \quad \mathbf{R}^k f_! G \hookrightarrow \mathbf{R}^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o,$$

$$(c') \quad \mathbf{R}^j \tilde{f}_! G \cong \mathbf{R}^j \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } j < k \text{ and } t \geq t_o,$$

$$(c'_k) \quad \mathbf{R}^k \tilde{f}_! G \hookrightarrow \mathbf{R}^k \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } t \geq t_o,$$

$$\begin{aligned} (d) \quad \text{SS}_k(\mathbf{R} f_! G) &\subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ &= f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\text{supp}(G)$ and $f = q \circ \tilde{f}$.

Proof. Let K be a compact subset of $X \times \mathbf{R}$. If one sets $K_1 := q(K)$ and $K_2 := \tilde{\varphi}(K)$, then $K \subset K_1 \times K_2$ and we have

$$\tilde{f}^{-1}(K) \subset \tilde{f}^{-1}(K_1 \times K_2).$$

Thus it suffices to show that $\tilde{f}^{-1}(K_1 \times K_2) \cap \text{supp } G$ is compact. Now we have $\tilde{f}^{-1}(K_1 \times K_2) = f^{-1}(K_1) \cap \varphi^{-1}(K_2)$. Since K_2 is a compact subset of $\mathbf{R}$, we may assume $K_2 = [a, b]$ for some $a \leq b$. In this case, it follows that

$$f^{-1}(K_1) \cap \varphi^{-1}([a, b]) \cap \text{supp } G \subset f^{-1}(K_1) \cap \varphi^{-1}(]-\infty, b]) \cap \text{supp } G$$

and by the hypothesis that $\overline{Y_b} \cap \text{supp } G$ is proper over X , $f^{-1}(K_1) \cap \overline{Y_b} \cap \text{supp } G$ is compact. Then the result follows. $\square$

Let us denote by i_t the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and by $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams, in both of which the left squares are cartesian.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X \\ \uparrow j_t & \square & \uparrow i_t \nearrow q_t \\ Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t[\end{array} \quad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X \\ \uparrow \tilde{j}_t & \square & \uparrow \tilde{i}_t \nearrow \tilde{q}_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t] \end{array}$$

Then, setting $\tilde{G} := Rf_*G$, we have

$$\begin{aligned} Rf_*G &\cong Rq_*\tilde{G}, & R\tilde{f}_*G &\cong R\tilde{q}_*\tilde{G}, \\ Rf_{t*}j_t^{-1}G &\cong Rq_{t*}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R\tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ Rf_{t!}j_t^{-1}G &\cong Rq_{t!}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^!G &\cong R\tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Thus taking j -th cohomology, we have

$$\begin{aligned} R^j f_*G &\cong R^j q_*\tilde{G}, & R^j \tilde{f}_*G &\cong R^j \tilde{q}_*\tilde{G}, \\ R^j f_{t*}j_t^{-1}G &\cong R^j q_{t*}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ R^j f_{t!}j_t^{-1}G &\cong R^j q_{t!}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^!G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .

(i) It follows from the hypothesis that

$$(2.1) \quad \mathrm{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any $(y; \eta) \in \mathrm{SS}_k(G)$ and $(f(x); \xi) \in Y \times_X T^*X$, we have

$$(2.2) \quad d\varphi(y) \neq \eta + f^*\xi.$$

If $(x, t; \xi, \tau)$ is a point of $\mathrm{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$, then, by the formula (1.1), there exists $(y; \eta) \in \mathrm{SS}_k(G)$ such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

For this η , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (2.2) because $\mathrm{SS}_k(G)$ is conic.

By (2.1), Let U be an open subset of X contained in a local chart, and convex in such a chart.

このとき, $U_t: U \times]-\infty, t[$ として (U_t) を考えても打切り版非特性変形補題は使えない。□

参考文献

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